

This Month's DVD Contents

This month's DVD is a collection of a variety of electronics software, ranging from the PCB design suite from Cadence to the semiconductor simulator for submicron and nanoscale devices. Make sure to try the free e-adventure for all your fun creations

SNEHA AMBASTHA

OrCAD Lite (version 16.6)

OrCAD Lite is the demo package of OrCAD design solutions for a PCB. From the initial schematic to the final artwork, you will find the complete tools for an integrated workflow. The set of tools included are OrCAD Capture, OrCAD CIS, PSpice A/D, PSpice advanced analysis, OrCAD PCB editor, SPECCTRA for OrCAD and OrCAD signal integrity, limited only by the size and complexity of the design. This is basically the full-functional version that offers all features of OrCAD. Unlike all other free and trial versions, OrCAD Lite never expires so you can keep using it.

GHD (version 0.31)

G hardware design language, popularly known as GHD, is a very high-speed integrated circuit hardware description language (VHDL) compiler. It converts VHDL codes directly to machine codes without interference from any intermediary languages, like C or C++. It can directly create binaries or executable images that are considered to be the best form for test benches because these are autonomous, self-checking VHDL designs. GHD is an open source software available for free.

Archimedes (version 2.0.1)

Archimedes is a simulator available for submicron and nanoscale semiconductor devices. It is available under the GNU general-public licence (GPL), and thus can be easily downloaded, used, modified and redistributed. It is very easy to set up and run a simulation with Archimedes, as you do not need to know the physics or the code behind it to use it

effectively. It is based on the Monte Carlo method of simulation.

OpenSCADA (version 1.3.0)

OpenSCADA is a full-featured, state-of-the-art supervisory control and data acquisition (SCADA) system that is used to control distributed systems used in industrial and civil engineering applications, from a master location. It is a free SCADA tool available under the GNU lesser general-public licence (LGPL). This particular release has many new features and enhancements to the existing functionality.

Circuit construction kit (DC only) (version 3.20)

It is a light-weight utility software to help create electronics circuits using different preset components. It has been mainly brought up for educational purposes to teach students about electrical components that are used in series and parallel circuits. It is a Java based portable program for Windows XP/Vista/7 OSes and does not leave traces in registry keys. A freeware, it can easily be copied on a USB and used on a computer with pre-installed Java environment.

DX Toolbox (version 4.4.0)

DX Toolbox is a short-wave propagation-forecasting tool. It searches the web, gathers information on solar and geomagnetic conditions that affect radio propagation. This is the free version of the software, available both for Windows and Mac OS X. It has several propagation tools associated with it that allow one to

Popular resources

- 7Zip. Version 9.35 (file archiver)
- Java Runtime Environment. Version 8.0.250.18 (runs programs written in Java programming language)
- VLC. Version 2.1.5 (media player)
- Opera. Version 24.0.1558.66 (web browser)
- Free Download Manager. Version 3.9.4 (file download manager)
- Thunderbird. Version 31.1.0 (email application)
- Mozilla Firefox. Version 34.0.5 (web browser)
- Apache Open Office. Version 4.1.1 (open office)
- Talisman Security (antivirus)

estimate current high-frequency (HF) short-wave propagation conditions between two locations of the world, quickly and easily.

Transformer Calculation (version 0.1)

It is a free software tool used to calculate the number of coils and wire thickness in the core of a transformer. It can be used only to calculate values for standard transformer cores with profile as either W or U, and does not support any toroid transformers with a ring profile core or any other type of transformers. It also helps identify primary and secondary coil parameters, like input and output voltage, output amperage, calculated power and lists of wanted power, amperage and voltage.

Toolduino (version 1.0)

Toolduino is a free software application that helps you work with Arduino hardware so that you can test circuits so created. It uses Arduino library for processing to communicate with any Arduino board. However, Toolduino cannot be used with any Arduino sketch; it requires a standard Firmata sketch on Arduino to work effectively. ●

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